Dawson S. Gaynor

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Education

Texas Tech University — Lubbock, TX
Incoming Ph.D student in Physics — *expected start: Fall 2026*

Carthage College — Kenosha, WI
B.S. Physics, Concentration in Astrophysics, Minor in English — *2022-2026*

- GPA: 3.72/4.0, Honors Program

Research Experience

Quantifying Point Source Recoverability within the Nanohertz Gravitational Wave Maps — Aug 2025 – Present
Mentored by Drs. Gosia Curyło, Eric Thrane, & Paul Lasky — *Monash University (Remote)*

- Created a tool to quantitatively compare and search for point sources within gravitational wave sky maps
- Simulated different gravitational wave sky maps based on different Pulsars and GW sources
- Tested code to ensure quality, reliability, and usability

NSF-International REU at Monash University — May 2025 – Aug 2025
Mentored by Drs. Gosia Curyło, Eric Thrane, & Paul Lasky — *Monash University*

- Familiarized with the typical Pulsar Timing Array analysis and simulation tools, and noted their failures
- Utilized a novel analysis tool to separate the simulated stochastic gravitational wave signal into four separate parts and create sky maps of the gravitational wave signal
- Created a tool to quantitatively compare maps and determine point source recovery

Gravitational Waves Research and Data Analysis — Aug 2024 – May 2025
Mentored by Dr. Jean Quashnock — *Carthage College*

- Used Bayesian inference with compact binary merger events gathered from the Gravitational Wave Open Science Center (GWOSC) to extract an event's source parameters
- Determined the final properties of events using GWOSC simulation tools with source parameters from Bayesian inference to determine the final properties of events
- Modeled the calculated mass/energy loss and compared it to older mass/energy loss models

Microgravity Ullage Detection Experiment — Aug 2024 – May 2025
Mentored by Dr. Kevin Crosby — *Carthage College Space Sciences*

- Worked on a student-led and NASA-sponsored team: Microgravity Ullage Detection (MUD), working to detect ullage bubbles in spacecraft propellant tanks
- Constructed three duplicates of the MUD payload to be used for the NASA Universal payload interface challenge, where three winning organizations will test their interface

Summer Undergraduate Research Experience (SURE) in Orientation Sensing for Dropsonde Probes — June 2024 – Aug 2024
Mentored by Dr. Brant Carlson — *Carthage College*

- Created an orientation sensing system for a free-falling dropsonde probe, which cheaply and accurately measures compass heading, frequency of rotation, and probe orientation
- Designed circuits with multiple types of operational amplifiers and micro-controllers to ensure accurate and easy-to-read results
- Simulated expected results to aid in the circuit building process, as well as ensuring the feasibility of the probe

Publications

M. Curyło, E. Thrane, P. D. Lasky, and **D. S. Gaynor**, *A comprehensive framework for phase-coherent mapping of the gravitational-wave sky with pulsar timing arrays, (2026) https://arxiv.org/abs/2604.19073*

Oral Presentations

Midstates Physical Sciences Research Symposium|University of Chicago — Nov 2025
Oral - "Point Source Recovery Using Phase-Coherent Gravitational-Wave Sky Maps"

University of Florida IREU Debrief|Albert Einstein Institute in Hannover, Germany — Aug 2025
Oral - "Point Source Recovery Using Phase-Coherent Gravitational-Wave Sky Maps"

Poster Presentations

Celebration of Scholars|Carthage College — April 2025
Poster - "Estimating the Gravitational-Wave Energy, Fractional Mass Loss, and Luminosity Distance of BBH Merger Events in the LIGO O3-O4 Data Catalogs"
Abstract available electronically at
https://app.carthage.edu/forms/scholars/archives/presentation/screen/2025/

Midstates Physical Sciences Research Symposium|Wash U in St. Louis — Nov 2024
Poster - "Orientation Sensing for Electric Field Sensors for Thunderstorms"
Abstract available electronically at
https://www.mathsciconsortium.org/undergraduate-symposia/

SURE Presentation|Carthage College — Sept 2024
Poster - "Orientation Sensing for Electric Field Sensors for Thunderstorms"

Extracurricular

Planetarium Co-President — Aug 2025 - Present
Conducted planetarium shows and open hours for the student population, and school guests. Trained younger students on astronomy and planetarium usage.

Carthage College Baseball — Aug 2022 - Present

Society of Physics Students — Aug 2022 - Present

Honors & Awards

Distinguished Graduate Student Assistantship |2026-Present

- Awarded by the Carthage physics faculty to one physics student (junior and below) per year for academic achievement in physics.

Outstanding Physics Student at Carthage College|2024-25

- Awarded by the Carthage physics faculty to one physics student (junior and below) per year for academic achievement in physics.

Carthage College Highest Honors Scholarship|2022-Present

- Scholarship awarded for academic success and accomplishments.

Carthage College Dean's List|2022 - 2025

- Students whose grade point average was 3.5 or above while earning at least 12 graded credits during the term.

Carthage College Honors Program

- Take specific courses offered to Honors students and obtain a total of fourteen Honors credits during their time at Carthage.

Spring CCIW Dave Wrath Academic All-Conference

- The Dave Wrath Academic All-Conference list comprises those individuals that achieved an overall grade-point average of 3.30 or above and have served at least one year in residency at their respective school.

State of Michigan Seal of Biliteracy: Spanish

- Recognize high school graduates who exhibit proficiency in English and at least one additional world language.

Burdick Thorne Scholarship|June 2022

- Awarded alongside the Kalamazoo Resa EFE/CTE Outstanding Senior Award in Aviation Technology.

Skills

Python, Jupyter, Matlab, Arduino, Julia, Slurm, Git, LaTeX, Inventor, Fusion 360, AstroImageJ, SAOImage, Excel